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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/674,113	02/18/2010	Ganga Raju Gokaraju	LIX 3026	4082

30868 7590 09/26/2011

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EXAMINER	
MCCORMICK, MELENIE LEE	

ART UNIT	PAPER NUMBER
1655	

NOTIFICATION DATE	DELIVERY MODE
09/26/2011	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

mail@krameramado.com

Office Action Summary	Application No.	Applicant(s)	
	12/674,113	GOKARAJU ET AL.	
	Examiner	Art Unit	
	MELENIE MCCORMICK	1655	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE *3* MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☒ Responsive to communication(s) filed on *07/15/2011*.

2a)☐ This action is **FINAL**. 2b)☒ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

5)☒ Claim(s) *1-4 and 16-30* is/are pending in the application.

5a) Of the above claim(s) *23,24 and 27* is/are withdrawn from consideration.

6)☐ Claim(s) _____ is/are allowed.

7)☒ Claim(s) *1-4, 16-22, 25-26, and 28-30* is/are rejected.

8)☐ Claim(s) _____ is/are objected to.

9)☐ Claim(s) _____ are subject to restriction and/or election requirement.

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☒ The drawing(s) filed on *18 February 2010* is/are: a)☒ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

12)☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

13)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

a)☐ All b)☐ Some * c)☐ None of:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948)

3) ☐ Information Disclosure Statement(s) (PTO/SB/08) Paper No(s)/Mail Date _____.

4) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date. _____.

5) ☐ Notice of Informal Patent Application

6) ☐ Other: _____.

DETAILED ACTION

Election/Restrictions

Applicant's election with traverse of group I, claims 1-4, 16-22, 25-26 and 28-30, in the reply filed on 0715/2011 is acknowledged. The traversal is on the grounds that groups I and II do in fact share a technical feature which defines a contribution over the prior art. In particular, Applicants argue that Parinitha (the art cited in the Restriction Requirement) describes use of *Holoptelea intefrolia* for treating hair loss/and or herpes. Applicants argue that their invention is directed to a composition which is a herbal supplement, which both contains *Holoptelea integrefolia* extract, and has the property of being anti-adapogenic and pro-lipolytic. Applicants further argue that Parinitha offers no evidence that compositions of *Holoptelea integrefolia* extract are anti-adapogenic and pro-lipolytic. This is not found persuasive because although Parinitha does not explicitly state this activity, the *Holoptelea integrefolia* extracts disclosed by would necessarily be anti-adapogenic and pro-lipolytic since the instant specification discloses that *Holoptelea integrifolia* extracts are useful as adipogenesis inhibitors and lipolysis accelerators (see e.g. pages 10 and 17). In addition, in light of the prior art references applied below, an anti-adipogenic and pro-lipolytic *Holoptelea integrifolia* extract was known at the time of the claimed invention.

The requirement is still deemed proper and is therefore made FINAL.

Claims 23-24 and 27 are withdrawn from further consideration pursuant to 37 CFR 1.142(b), as being drawn to a nonelected invention, there being no allowable generic or linking claim.

Claims 1-4, 16-22, 25-26, and 28-30 are presented for examination on the merits.

Information Disclosure Statement

The Information Disclosure Statements submitted 11/22/2010 and 05/20/2010 have been received and considered.

Document 1 and 3-4 on the IDS submitted 05/20/2010 is not considered because dates have not been provided for these documents.

Claim Objections

Claims 1-4, 16-22, 25-26, and 28-30 are objected to because of the following informalities: The claims are drawn to 'anti adipogenic and pro-lipolytic supplements'. It is suggested that the claim be amended to recite 'supplement' in order to more closely reflect conventional claim language.

Claim Rejections - 35 USC § 112

The following is a quotation of the first paragraph of 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same and shall set forth the best mode contemplated by the inventor of carrying out his invention.

Claims 1-4, 16-22, 25-26 and 28-30 are rejected under 35 U.S.C. 112, first paragraph, because the specification, while being enabling for an anti-adipogenic and pro-lipolytic herbal supplement for inhibiting or controlling the particular adipogenesis and lipolysis involved disease of obesity, overweight, and hyperlipidemia in mammals comprising a water or polar organic solvent extract or purified fraction derived from *Holoptelea integrifolia* alone or a composition comprising said water or polar organic solvent extract or fraction optionally in combination with at least one antiobesic agent does not does not reasonably provide enablement for an anti-adipogenic and pro-lipolytic herbal supplement for inhibiting or controlling or preventing all adipogenesis and lipolysis involved diseases in mammals comprising any extract or purified fraction derived from *Holoptelea integrifolia* alone or a composition comprising said extract in combination with at least one antiobesic agent. The specification does not enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the invention commensurate in scope with these claims.

The factors to be considered in determining whether a disclosure meets the enablement requirements of 35 U.S.C. 112, first paragraph, have been described in *In re Wands*, 858 F.2d 731, 8 USPQ2d 1400 (Fed. Cir., 1988). The court in Wands states, "Enablement is not precluded by the necessity for some experimentation, such as routine screening. However, experimentation needed to practice the invention must not be undue experimentation. The key word is 'undue', not 'experimentation'" (Wands, 8 USPQ2sd 1404). Clearly, enablement of a claimed invention cannot be predicated on the basis of quantity of experimentation required to make or use the invention. "Whether undue experimentation is needed is not a single, simple factual determination, but rather is a conclusion reached by weighing many factual considerations" (Wands, 8 USPQ2d 1404). Among these factors are: (1) the nature of the invention; (2) the breadth of the claims; (3) the state of the prior art; (4) the predictability or unpredictability of the art; (5) the relative skill of those in the art; (6) the amount of direction or guidance presented; (7) the presence or absence of working examples; and (8) the quantity of experimentation necessary.

While all of these factors are considered, a sufficient amount for a *prima facie* case is discussed below.

(1) The nature of the invention and (2) the breadth of the claims:

The claims are drawn to anti-adipogenic and pro-lipolytic herbal supplements for inhibiting or preventing or controlling adipogenesis and lipolysis involved diseases in mammals, comprising an extract or purified fraction derived from *Holoptelea integrifolia* alone or a composition comprising said extract or fraction optionally in combination with at least one antiobesic agent (see e.g. claim 1). The claims are also drawn to the supplement wherein said adipogenesis and lipolysis involved diseases comprise obesity, overweight, hyperlipidemia, type 2 diabetes, cardiovascular disease and atherosclerosis (see e.g. claim 4). Thus, the claims taken together with the specification imply that Applicants are claiming, in the broadest claim, a composition for inhibiting or controlling or preventing any disease in which adipogenesis or lipolysis is involved. In addition, the claims taken together with the specification also imply that Applicants are claiming a composition for inhibiting or controlling or preventing the specific conditions recited in instant claim 4. As discussed by labtestonline.org, cardiovascular disease is a general term used to describe disorders that can affect the heart and blood vessels see e.g. page 1) and includes cerebral vascular diseases such as stroke (see e.g. page 1) as well as peripheral vascular disease including gangrene (see e.g. pages 1-2) and myocarditis and congenital heart disease (see e.g. page 2). Therefore, the claims are broad in that they encompass numerous diseases. In addition, the claims are drawn to an anti-adipogenic and pro-lipolytic herbal supplement comprising an extract or purified fraction derived from *Holoptelea integrifolia*. Claim 20 is drawn to the extract wherein it is prepared using water, a polar organic solvent or a mixture of the two. Claim 21 is drawn to the extract wherein it is prepared using particular polar organic solvents as well as particular non-polar organic solvents. Therefore, the claims encompass an anti-adipogenic and pro-lipolytic supplement which comprises any type *of Holoptelea integrifolia* extract, including water extracts, and polar and non-polar organic solvent extracts.

(3) The state of the prior art and (4) the predictability or unpredictability of the art:

The state of the art is unpredictable with regard to prevention, inhibition and control of many of the conditions encompassed by the instant claims and it would be unpredictable determine which of the vast number of conditions encompassed by the instant claims could be controlled, prevented or inhibited with the instantly claimed anti-adipogentic and pro-lipolytic supplement. For instance, adipogenesis contributes to the pathogenesis of thyroid eye disease (see e.g. Crisp et al. page 3249, abstract), therefore, thyroid eye disease is an adipogenesis involved disease and is therefore encompassed by the instant claims. However, thyroid eye disease is an autoimmune diseases and the pathogenesis of thyroid eye disease is not completely clear (see e.g. Crisp et al. page 3249). In addition, as discussed by Camm, atherosclerosis is a chronic and multifocal immune-inflammatory, fibroproliferative disease of medium sized and large arteries driven by lipid accumulation and the speed of its progression is unpredictable and markedly different in different subjects (see e.g. page 338). Camm further teaches that at every level of risk factor exposure, there is substantial variation in the amount of evolved atherosclerosis probably because the individual vulnerability to atherosclerosis and its risk factors varies greatly (see e.g. page 338). In addition, while adipogenesis is involved in obesity, as discussed by Rosen et al., inhibition of adipogenesis alone is an inappropriate approach to anti-obesity therapy (see e.g. page 893 Box 3). Therefore, since many of the diseases encompassed by the claims are unpredictable or have an unclear pathology, determining which of the diseases encompassed by ‘adipogenesis and lipolysis involved diseases’ could be controlled or inhibited or prevented using the instantly claimed supplement would be unpredictable.

In addition, the state of the art is unpredictable with regard to plant extracts. It is well known in the art that polarity of solvents plays a key role in determining the final product obtained by an extraction. However, because many phytochemicals remain undiscovered, the skilled artisan has to make his best educated guess as to what types of phytochemicals will be successfully extracted with a solvent of a particular polarity. Often times, unless the constituents in a particular plant extract have been well evaluated and documented in the literature, the skilled artisan must adhere to trial and error protocols in order to quantitatively determine phytochemical constituents present in samples obtained from respective extraction procedures. These procedures are common when, for example, a plant or part thereof has been documented in the literature as possessing some medicinal quality. The skilled artisan will attempt numerous extraction protocols in attempt to isolate the particular ingredient which has this medicinal quality. Typically, beginning with the first crude extraction, it is a guess as to whether or not the extract will possess certain phytochemical constituents. It is noted that the Instant specification does not disclose what the active ingredient of the extract is; on the contrary, the specification only teaches certain extracts (i.e. methanol/polar) which provide for the effective ingredient.

Each successive extraction of plant matter yields different products due to the exclusion of ingredients based on the polarity of the solvents solvating constituents with similar polarities. Subsequently, the properties of each respective product are unpredictable and would need to be evaluated for chemical constituents.

Therefore, preparing an anti-adipogenic and pro-lipolytic composition comprising any type of extract of Holoptelia integrifolia would be unpredictable.

(5) The relative skill of those in the art:

The relative skill of those in the art is high.

(6) The amount of direction or guidance presented and (7) the presence or absence of working examples:

The specification has provided an example wherein an extract of *Holoptelea integrifolia* inhibited fat accumulation in differentiating adipocytes (see e.g. pages 14-15, Example 5). Therefore, the specification has reasonably demonstrated that the extract is anti-adipogenic. In addition, the specification has provided an example wherein an extract of Holoptelea increased lipolysis in adipocytes (see page 15-16, Example 6). Therefore, the specification has reasonably demonstrated that the extract is pro-lipolytic. The specification has also provided an example wherein rats fed a high fat diet and simultaneously fed the extract were afforded a 15.4% and 27.3% protection against weight gain (compared to the control group rats) with different doses of the extract (see page 16, Example 7). The rats treated with the extract also showed significant reduction in serum triglycerides and lipid profile and a reduction in serum cholesterol and LDL. The specification also provides an example wherein obesity was induced in rats and that the rats were then administered a *Holoptelea integrifolia* extract (see e.g. pages 16-17). The treated rats were afforded a reduction in weight gain of 98.1% and 130% compared to the rats in the control group and showed a significant reduction in serum triglycerides and lipid profile. Therefore, the specification reasonably demonstrates that the instantly claimed supplement is useful in inhibiting or controlling obesity or overweight and hyperlipidemia. However, the specification does not disclose that the extract actually prevented any diseases or conditions, including those claimed and encompassed by the instant claims. In addition, the specification does not disclose that a *Holoptelea integrifolia* extract inhibited or controlled or prevented any of the other numerous conditions instantly claimed and encompassed by the claims.

The specification has provided examples wherein methanol extracts, water extracts and water/methyl alcohol extracts of Holoptelea integrifolia were produced (see pages 13-14) and the methanol extract was evaluated for anti-adipogenic and pro-lipolytic activity, as discussed above. However, the specification does not disclose that any non-polar extracts were produced and that they had the activity instantly claimed.

In addition, it should be noted that Singh et al. disclose that a petroleum ether (i.e. non-polar) extract of the seeds of *Holoptelea integrifolia* contains, among other compounds, ß-sitosterol (see e.g. abstract). As evidenced by Chai et al., ß-sitosterol has been shown to stimulate adipogenesis (see e.g. abstract). Therefore, anti-adipogenic activity of a non-polar extract of *Holoptelea integrifolia*, due to the presence of a pro-adipogenic compound, would be unpredictable.

(8) The quantity of experimentation necessary:

Considering the state of the art as discussed by Rosen regarding the fact that inhibiting of adipogenesis alone is inappropriate approach to anti-obesity therapy, a person of ordinary skill in the art would be burdened with undue experimentation in order to determine the how to use the instantly claimed supplement to prevent obesity. In addition, considering the state of the art discussed by Crisp regarding the fact that the pathogenesis of thyroid eye disease is unclear and the lack of guidance provided in the specification with regard to control, inhibition or prevention of this disease with the instantly claimed extract , one of ordinary skill in the art would be burdened with undue experimentation to determine all of the patient populations, effective amounts and administration routes necessary in order to evaluate the extract for the prevention, control or inhibition of thyroid eye disease. Similarly, in light of the unpredictable nature of atherosclerosis as discussed by Camm and the lack of guidance in the instant specification with regard to the use of the claimed extract for prevention, inhibition or control of atherosclerosis, one of ordinary skill in the art would be burdened with undue experimentation to determine all of the patient populations, effective amounts and administration routes necessary in order to evaluate the extract for the prevention, control or inhibition of atherosclerosis. Likewise, since the instant specification does not provide guidance with respect to the use of the instantly claimed supplement for treatment of cardiovascular disease (and all of the conditions encompassed by this term, as discussed above), type 2 diabetes or any additional diseases in which adipogenesis and lipolysis would be related, it would take undue experimentation in order to use the instantly claimed supplement commensurate with the full scope of the claims. In addition, in light of the unpredictability with regard to the particular components extracted by various solvents, it would be unpredictable to prepare an anti-adipogenic and pro-lipolytic herbal supplement comprising an extract of *Holoptelea integrifolia* for use in inhibiting or controlling adipogenesis and lipolysis involved diseases wherein the extract is obtained using any solvent.

Please note that this rejection can be overcome by omitting the phrase 'for inhibiting or preventing or controlling adipogenesis and lipolysis involved diseases in mammals' from claim 1 and rewriting claim 4 as an independent claim which is drawn to the composition in claim 1 for inhibiting or controlling adipogenesis and lipolysis involved disease wherein the diseases are selected from the group consisting of obesity, overweight and hyperlipidemia. In addition, the rejection can be overcome by, in addition to the previous suggestion, limiting the extract claimed (*Holoptelea integrifolia*) to a water or polar organic solvent extract.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (b) the invention was patented or described in a printed publication in this or a foreign country or in public use or on sale in this country, more than one year prior to the date of application for patent in the United States.

Claims 1, 3-4, 16, 18-19, and 25-26 are rejected under 35 U.S.C. 102(b) as being anticipated by Sodhala et al. (1999) (English Translation) with evidence provided by Anwikar et al. (2010).

Sodhala et al. teach a therapeutic composition which is made by powdering plant material including the fruit of *Holoptelea integrifolia* and the fruit or *Solanum xanthocarpum*, boiling the powdered plant material with water, wherein a specific quantity of water is retained after boiling and then filtering the composition to obtain the decoction (Kvatha) (see e.g. pages 5-6 of 15). Therefore, Sodhala et al. teach a decoction (which is a water extract) of herbs, including *Holoptelea integrifolia* and *Solanum xanthocarpum*. Since the composition contains water, the composition contains a biologically acceptable excipient and therefore is formulated as a nutraceutical or pharmaceutical dosage form, as instantly claimed. In addition, nothing would preclude the parenteral administration of the decoction disclosed by Sodhala et al. and the composition could be injected or used in drop form since it is a liquid containing water as an excipient. Furthermore, Sodhala et al. teach that the composition is orally administered and is a therapeutic composition and is useful in the treatment of, among other conditions, obesity (see page 6 of 15) therefore, the composition reads on a food for specified health uses, as instantly claimed.

Anwikar et al. teach that a water extract of the fruits of *Solanum xanthocarpum* has anti-inflammatory activity (see e.g. page 1). Therefore, since the composition disclosed by Sodhala et al. contains a decoction (i.e. an aqueous extract) of *Solanum xanthocarpum*, the composition contains a pharmaceutically, nutritionally or dietetically acceptable anti-inflammatory agent, as instantly claimed. Nothing would preclude the administration of the composition, which reads on a food or beverage as instantly claimed (since it is orally administrable) in an amount of 0.01 to 300 mg/kg per day. It should be noted that instant claim 28 recites the manner in which the composition is used, rather than a unit dosage form of the composition. In addition, since the composition meets the structural limitations of the instant claims (i.e. it contains a water extract from *Holoptelea integrifolia*), it is an anti-adipogenic and pro-lipolytic herbal supplement, as instantly claimed. Therefore, nothing would preclude its use as such, as instantly claimed. Furthermore, nothing would preclude the use of the composition disclosed by Sodhala et al. for inhibiting or controlling the adipogenesis and lipolysis involved disease of obesity, overweight, and hyperlipidemia since it meets the structural limitations of the instant claims. In addition, as discussed above, Sodhala et al. explicitly disclose that the composition is useful in treating obesity (see page 6 of 15).

Therefore, the reference is deemed to anticipate the instant claims above.

Claims 1-4, 16, 19, 26, and 28-29 rejected under 35 U.S.C. 102(b) as being anticipated by Sharma et al. (1994).

Sharma et al. teach that an aqueous extract made from the seeds of *Holoptelea integrifolia* (see e.g. page 1). Sharma et al. further teach that the extract was made by grinding 2 grams of the seeds with a mortar and pestle in distilled water (see e.g. page 1). Therefore, since the composition disclosed by Sharma et al. contains a water extract of the seeds of *Holoptelea integrifolia* as instantly claimed, the composition is an anti-adipogenic and pro-lipolytic supplement as instantly claimed and nothing would preclude its use as an anti-adipogenic and pro-lipolytic supplement for inhibiting or controlling the adipogenesis and lipolysis involved disease of obesity, overweight, and hyperlipidemia since it meets the structural limitations of the instant claims. Since the composition contains distilled water, it contains a biologically acceptable excipient and is formulated into a nutraceutical or pharmaceutical dosage form, as instantly claimed. In addition, nothing would one from eating the composition disclosed by Sharma et al. Therefore, it reads on a dietary formulation in the form of food. The use of the composition disclosed by Sharma et al. for specified health uses would not be precluded since it meets the structural limitations of the instant claims. In addition, nothing would preclude one from administering the composition in the form of a therapeutic formulation or food in a range of from 0.01 to 300 mg/kg/day. It should be noted that instant claim 28 recites the manner in which the composition is used, rather than a unit dosage form of the composition.

Therefore, the reference is deemed to anticipate the instant claims above.

Claim Rejections - 35 USC § 102/103

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (b) the invention was patented or described in a printed publication in this or a foreign country or in public use or on sale in this country, more than one year prior to the date of application for patent in the United States.

The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

> (a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negatived by the manner in which the invention was made.

Claims 1, 3-4, 16, 18-20 and 25-26 are rejected under 35 U.S.C. 102(b) as anticipated by or, in the alternative, under 35 U.S.C. 103(a) as obvious over by Sodhala et al. (1999) (English Translation) with evidence provided by Anwikar et al. (2010).

Sodhala et al. teach a therapeutic composition which is made by powdering plant material including the fruit of *Holoptelea integrifolia*, boiling the powdered plant material with water, wherein a specific quantity of water is retained after boiling and then filtering the composition to obtain the decoction (Kvatha) (see e.g. pages 5-6 of 15) and is relied upon for the reasons set forth above. While Sodhala et al. do not explicitly teach that the filtration step is performed with fine filters, that the filtrate is evaporated after filtration step and then purified, it should be noted that the extract of instant claim 20 is part of a composition which may comprise additional ingredients. Therefore, although the solvent (which may be water) has been evaporated, when adding the concentrated extract to an herbal composition which may comprise additional ingredients, including water, the extract would not be expected to differ from an extract from which water had not been evaporated. In addition, while instant claim 20 recites the steps of using fine filter and of purifying, it is not explicitly disclosed how fine the filters are or how the purification takes place or to what degree the extract is purified. Therefore, absent evidence to the contrary, the extract disclosed by Sodhala, which has also been filtered, would be expected to be substantially the same as the instantly claimed extract.

With respect to the USC 102/103 rejection above, please note that the patentability of a product does not depend on its method of production. If the product in the product-by-process claim is the same as or obvious from a product of the prior art, the claim is unpatentable even though the prior product was made by a different process." In re Thorpe, 777F.2d 695, 698, 227 USPQ 964, 966 (Fed. Cir. 1985) (citations omitted). In addition, please also note that "The Patent Office bears a lesser burden of proof in making out a case of prima facie obviousness for product-by-process claims because of their peculiar nature" than when a product is claimed in the conventional fashion. In re Fessmann, 489 F.2d 742, 744, 180 USPQ 324, 326 (CCPA 1974). Once the examiner provides a rationale tending to show that the claimed product appears to be the same or similar to that of the prior art, although produced by a different process, the burden shifts to applicant to come forward with evidence establishing an unobvious difference between the claimed product and the prior art product. In re Marosi, 710 F.2d 798, 802, 218 USPQ 289, 292 (Fed. Cir. 1983) (See MPEP 2113).

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

> (a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negatived by the manner in which the invention was made.

Claims 1, 3-4, 16-20, 25-26 and 30 are rejected under 35 U.S.C. 103(a) as being unpatentable over Sodhala et al. (1999) (English Translation) with evidence provided by Anwikar et al. (2010) Jacobs et al. (US 2005/0130933 A1).

Sodhala et al. teach a therapeutic composition which comprises a water extract of the fruit of *Holoptelea integrifolia* and a biologically acceptable excipient (i.e. water) which reads on an anti-adipogenic and pro-lipolytic herbal supplement, as instantly claimed, and are relied upon for the reasons set forth above. As discussed above, Sodhala et al. teach that the composition is orally administered and is a therapeutic composition and is useful in the treatment of, among other conditions, obesity (see page 6 of 15).

Sodhala et al. do not explicitly teach that the composition is formulated as a tablet, capsule or nutritional food bar.

Jacobs et al. teach a method for treating weight loss by administering an herbal composition (see e.g. claim 34). Jacobs et al. teach that the composition may be administered in the form of a capsule (which would necessarily either be hard or soft), a tablet or a food bar (see e.g. claims 47, 48 and 50).

It would have been obvious to one of ordinary skill in the art to formulate the herbal composition for treating obesity disclosed by Sodhala et al. in the form of a tablet, capsule or nutritional food bar. A person of ordinary skill in the art would have had a reasonable expectation of success in doing so since Sodhala et al. teach that the composition is orally administered and since Jacobs et al. teach that another herbal composition for weight loss may be formulated as a tablet, capsule, or food bar. Therefore, a person of ordinary skill in the art would reasonably recognize that these are conventional forms for herbal compositions which are used for weight loss. Since the food bar would contain at least some food, it would be nutritional.

From the teachings of the references, it is apparent that one of ordinary skill in the art would have had a reasonable expectation of success in producing the claimed invention. Therefore, the invention as a whole was *prima facie* obvious to one of ordinary skill in the art at the time the invention was made, as evidenced by the references, especially in the absence of evidence to the contrary.

Claims 1, 3-4, 16, 18-21 and 25-26 are rejected under 35 U.S.C. 103(a) as being unpatentable over by Sodhala et al. (1999) (English Translation) with evidence provided by Anwikar et al. (2010) in view of Mason (1999).

Sodhala et al. teach a therapeutic composition which comprises a filtered decoction of the fruit of *Holoptelea integrifolia* and a biologically acceptable excipient (i.e. water) which would be expected to be substantially the same as the instantly claimed composition although it was made in a slightly different manner and are relied upon for the reasons set forth above.

Sodhala et al. do not explicitly teach that the *Holoptelea integrifolia* is extracted with ethanol.

Mason teaches that preparations made with herbs usually require the addition of a base solvent such as water or alcohol. Mason further teaches that aqueous preparations such as infusions and decoctions do not keep very well, but that alcohol – based tinctures have a much longer shelf like. Mason further teaches that decoctions have another advantage in that alcohol is a very good solvent for many constituents in plants. Mason further teaches an example of a plant extraction wherein ethanol proved to be a better solvent than aqueous ethanol.

A person of ordinary skill in the art would have had a reasonable expectation of success in performing the extraction disclosed by Sodhala et al. with ethanol instead of water. A person of ordinary skill in the art would have had a reasonable expectation of success in doing so since Mason teaches that decoctions do not keep very well and that in alcohol based extracts have a longer shelf life. Combined with the additional disclosure of Mason that ethanol may be a superior extraction solvent compared to aqueous ethanol, a person of ordinary skill in the art would have good reason to prepare the extract disclosed by Sodhala et al. using ethanol.

From the teachings of the references, it is apparent that one of ordinary skill in the art would have had a reasonable expectation of success in producing the claimed invention. Therefore, the invention as a whole was *prima facie* obvious to one of ordinary skill in the art at the time the invention was made, as evidenced by the references, especially in the absence of evidence to the contrary.

Claims 1, 3-4, 16, 18-20, 22, and 25-26 are rejected under 35 U.S.C. 103(a) as being unpatentable over Sodhala et al. (1999) (English Translation) with evidence provided by Anwikar et al. (2010) in view of Watanabe (2004).

Sodhala et al. teach a therapeutic composition which comprises a water extract of the fruit of *Holoptelea integrifolia* and a biologically acceptable excipient (i.e. water) which reads on an anti-adipogenic and pro-lipolytic herbal supplement, as instantly claimed, and are relied upon for the reasons set forth above. As discussed above, Sodhala et al. teach that the composition is orally administered and is a therapeutic composition and is useful in the treatment of, among other conditions, obesity (see page 6 of 15).

Sodhala et al. do not explicitly teach that the composition additionally comprises a green tea extract or banaba extract.

Watanabe teaches that there are several therapeutic interventions by natural products in the treatment of obesity including green tea and banaba extract (see e.g. page 195).

It would have been obvious to one of ordinary skill in the art at the time the claimed invention was made to add green tea extract and/or banaba extract to the composition disclosed by Sodhala et al. A person of ordinary skill in the art would have had a reasonable expectation of success in doing so since Sodhala et al. teach that the composition is used for treating obesity and since Watanabe teaches that green tea and banana extract are also useful in treating obesity. Therefore, a person of ordinary skill in the art, when preparing the herbal composition disclosed by Sodhala et al., would have good reason to add green tea extract and/or banaba extract to the composition in order to prepare a natural treatment for obesity.

"The idea for combining them flows logically from their having been used individually in the prior art"; *In re Kerkhoven*, 626 F.2d 846, 850, 205 USPQ 1069, 1072 (CCPA 1980) (citations omitted) (Claims to a process of preparing a spray-dried detergent by mixing together two conventional spray-dried detergents were held to be *prima facie* obvious.). This rejection is based upon the well- established proposition of patent law that no invention resides in combining old ingredients of known properties where the results obtained thereby are no more than the additive effect of the ingredients, See *In re Sussman*, 136 F.2d 715, 718, 58 USPQ 262, 264 (CCPA 1943).

From the teachings of the references, it is apparent that one of ordinary skill in the art would have had a reasonable expectation of success in producing the claimed invention. Therefore, the invention as a whole was *prima facie* obvious to one of ordinary skill in the art at the time the invention was made, as evidenced by the references, especially in the absence of evidence to the contrary.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory obviousness-type double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the conflicting application or patent either is shown to be commonly owned with this application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement.

Effective January 1, 1994, a registered attorney or agent of record may sign a terminal disclaimer. A terminal disclaimer signed by the assignee must fully comply with 37 CFR 3.73(b).

Claims 1, 3-4, 16-19, 22, 25-26 and 30 are provisionally rejected on the ground of nonstatutory obviousness-type double patenting as being unpatentable over claims 23-41 and 45-51 of copending Application No. 12/678, 826.

This is a provisional obviousness-type double patenting rejection.

The claims of ‘826 are drawn to an anti-adipogenic herbal supplement composition for inhibiting or preventing or ameliorating disease conditions in mammals which comprises an anti adipogenic or anti-obese agent selected from a group with includes *Holoptelea integrifolia* and which may comprise a biologically acceptable excipient (see e.g. claims 30 and 45). The composition may also comprise many of the anti-obesic agents claimed in instant claim 22 (see claim 30 of ‘826). In addition, the claims of ‘826 are further drawn to an anti-adipogenic composition further comprising the agents claimed in instant claim 25 (see e.g. claim 32 of ‘826). In addition, the claims of ‘826 are drawn to an anti-adipogenic composition in the form of injections, drops and suppositories (see e.g. claim 34 of ‘826) and the forms claimed in instant claim 17 (see e.g. claim 33 of ‘826). The composition of ‘826 may also be in the dietary formulations claim in instant claim 30 (see claim 48 of ‘826), which read on foods for health use as claimed in instant claim 19.

While the claims of ‘826 do not disclose a particular embodiment wherein the anti-adipogenic composition contains the anti-adipogenic extract of *Holoptelea integrifolia* and is in the forms instantly claimed, a person of ordinary skill in the art would reasonably understand that an anti-adipogenic composition comprising *Holoptelea integrifolia* could be formulated in these ways since the claims of ‘826 are drawn to anti-adipogenic compositions formulated in these ways. In addition, since the composition rendered obvious by the claims of ‘826 contains an anti-adipogenic extract of *Holoptelea integrifolia* and would also contain the additional anti-obesic agents instantly claimed, it would be expected to have anti-adipogenic and anti-lipolyticc activity as instantly claimed because it would be expected to contain the same extract as instantly claimed. Furthermore, nothing would preclude the use of the composition rendered obvious by the claims of '826 from being used in the manner instantly claimed since it would meet the structural limitations of the instant claims. Similarly, nothing would preclude the composition, including the food and beverage compositions, rendered obvious by the claims of'826 from being administered in an amount of 0.01 to 300 mg/kg weight/day.

Claims 1, 3-4, 16-19, 22, 25-26 and 30 are provisionally rejected on the ground of nonstatutory obviousness-type double patenting as being unpatentable over claims 1-16, 18, 20, 22, 24-26, 31-32, 34 and 38-39 of copending Application No. 12/784,295 in view of Jacobs et al. (US 2005/0130933 A1)

The claims of '295 are drawn to an anti-adipogenic and pro-lipolytic composition for amelioration of adipogenesis and lipolysis related diseases which may comprise a *Holoptelea integrifolia* extract as an anti-obesic agent (see e.g. claim 12). In addition, the anti-adipogenic and pro-lipolytic composition of ‘295 may also comprise many of the anti-obesic agents as claimed in instant claim 22 (see claim 12 of ‘295). The composition of ‘295 may also biologically acceptable carriers (see e.g. claim 2) and at least one pharmaceutically or nutritionally of dietetically acceptable antioxidant, adaptogen, anti-inflammatory agent, anti-diabetic agent, or trace element (see e.g. claim 14). In addition, the composition of ‘295 may also be in the form of a dietary formulation as a food for specific health uses (see claim 15) and may be formulated as many of the forms recited in instant claim 30 (see e.g. claim 34 of '295). The claims of '295 are also drawn to the composition wherein it is formulated as a pharmaceutically or dietary or nutraceutical formulation (see claim 24).

The claims of ‘295 are not explicitly drawn to an anti-adipogenic and pro-lipolytic composition for amelioration of adipogenesis and lipolysis related diseases comprises a *Holoptelea integrifolia* extract which is in the particular forms instantly claimed, including a tablet and capsule.

Jacobs et al. teach a method for treating weight loss by administering an herbal composition (see e.g. claim 34). Jacobs et al. teach that the composition may be administered in the form of a capsule (which would necessarily either be hard or soft) or a tablet (see e.g. claims 47and 48).

It would have been obvious to one of ordinary skill in the art to prepare a particular anti-adipogenic and pro-lipolytic composition comprising an extract of *Holoptelea integrifolia* since the claims of ‘295 include *Holoptelea integrifolia* as an anti-obesic agent which may be included in the formulation. In addition, it would have been obvious to one of ordinary skill in the art to formulate the anti-adipogenic and pro-lipolytic composition comprising a *Holoptelea integrifolia* extract rendered obvious by the claims of '295 as a tablet or capsule. A person of ordinary skill in the art would have had a reasonable expectation of success in doing so since the claims of ‘295 disclose that the composition is in the form of a pharmaceutical or dietary or nutraceutical composition (see claim 24) and since Jacobs et al. teach that another herbal composition for weight loss may be formulated as a tablet or capsule. Therefore, a person of ordinary skill in the art would reasonably recognize that these are conventional forms for herbal compositions which are used for weight loss. Likewise, a person of ordinary skill in the art would be motivated to formulate an anti-adipogenic and pro-lipolytic composition in a form as recited in instant claim 30 since the claims of ‘295 are drawn to an anti-adipogenic and pro-lipolytic agent which may be formulated in these ways (see e.g. claim 34 of '295).

Furthermore, nothing would preclude the use of the composition rendered obvious by the claims of '295 from being used in the manner instantly claimed since it would meet the structural limitations of the instant claims. Similarly, nothing would preclude the composition, including the food and beverage compositions, rendered obvious by the claims of '295 from being administered in an amount of 0.01 to 300 mg/kg weight/day.

This is a provisional obviousness-type double patenting rejection because the conflicting claims have not in fact been patented.

Conclusion

No claim is allowed.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to MELENIE MCCORMICK whose telephone number is (571)272-8037. The examiner can normally be reached on M-F 7:30-4:00.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Terry McKelvey can be reached on 571-272-0775. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Melenie McCormick/
Primary Examiner, Art Unit 1655

Notice of References Cited	Application/Control No. 12/674,113	Applicant(s)/Patent Under Reexamination GOKARAJU ET AL.	
	Examiner MELENIE MCCORMICK	Art Unit 1655	Page 1 of 3

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*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-2005/0130933	06-2005	Jacobs et al.	514/054
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	C	US-			
	D	US-			
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	X	Rosen et al. Adipocyte Differentiation from the Inside Out. Nature Reviews Molecular Cell Biology. Volume 7, December 2006. Pages 885-896.

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
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*	V	Sodhala et al. Indradeva Tripathi. part 2 (Kaya cikitsa Khanda Cahukhamba Sanskrit Sanstan (Varanasi) 3rd Edition. 1999, Page 653, Translation.
	W	Anwikar et al. Study of the Synergistic Anti-Inflammatory Activity of Solanum xanthocarpum Schrad and Wendl and Cassia Fistula Linn. Int. J Ayurveda Res. 2010. Jul-Sep 1 (2) : 167-171. (pages 1-7 printed).
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	V	Watanabe. Proceedings of the 3rd International Conference on Food Factors. IOS Press. 2004. page 195.
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